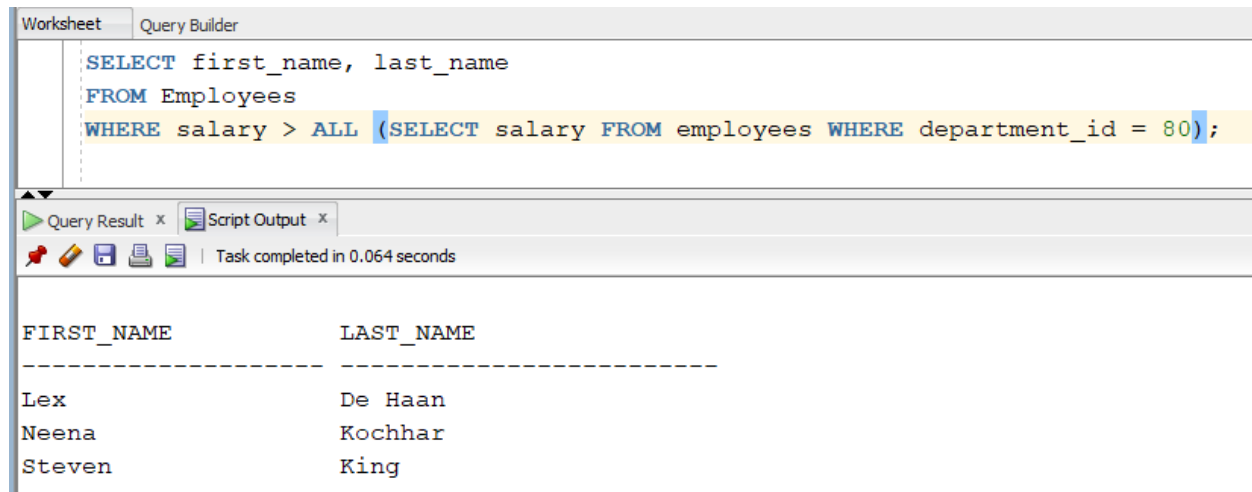


23K-2001 BCS-5J

DB LAB 4

Question 1:



The screenshot displays a database query builder interface. At the top, there are two tabs: "Worksheet" and "Query Builder". The "Query Builder" tab is active, showing a SQL query in a text area. The query is: `SELECT first_name, last_name FROM Employees WHERE salary > ALL (SELECT salary FROM employees WHERE department_id = 80);`. Below the query area, there is a toolbar with icons for running the query, saving, and other functions. A status bar indicates "Task completed in 0.064 seconds". Below the toolbar, the query results are displayed in a table with two columns: "FIRST_NAME" and "LAST_NAME". The results are as follows:

FIRST_NAME	LAST_NAME
Lex	De Haan
Neena	Kochhar
Steven	King

Question 2:

Worksheet Query Builder

```
SELECT first_name, last_name, hire_date
FROM Employees
WHERE hire_date < (SELECT hire_date FROM employees WHERE employee_id = 150);
```

Query Result x Script Output x

Task completed in 0.096 seconds

FIRST_NAME	LAST_NAME	HIRE_DATE
Steven	King	17-JUN-03
Lex	De Haan	13-JAN-01
Nancy	Greenberg	17-AUG-02
Daniel	Faviet	16-AUG-02
Den	Raphaely	07-DEC-02
Alexander	Khoo	18-MAY-03
Matthew	Weiss	18-JUL-04
Payam	Kaufling	01-MAY-03
John	Russell	01-OCT-04
Karen	Partners	05-JAN-05
Janette	King	30-JAN-04

FIRST_NAME	LAST_NAME	HIRE_DATE
Patrick	Sully	04-MAR-04
Allan	McEwen	01-AUG-04
Ellen	Abel	11-MAY-04
Jennifer	Whalen	17-SEP-03
Michael	Hartstein	17-FEB-04
Susan	Mavris	07-JUN-02
Hermann	Baer	07-JUN-02
Shelley	Higgins	07-JUN-02
William	Gietz	07-JUN-02

20 rows selected.

Question 3:

Worksheet Query Builder

```
SELECT j.job_title
FROM Employees e
JOIN jobs j ON e.job_id = j.job_id
WHERE e.salary = (SELECT MAX(salary) FROM Employees);
```

Query Result x

SQL | All Rows Fetched: 1 in 0.005 seconds

JOB_TITLE
1 President

Question 4:

Worksheet Query Builder

```
SELECT first_name, last_name
FROM Employees
WHERE department_id = (SELECT department_id FROM Employees WHERE first_name = 'Neena');
```

Script Output x Query Result x

SQL | All Rows Fetched: 3 in 0.004 seconds

FIRST_NAME	LAST_NAME
1 Steven	King
2 Neena	Kochhar
3 Lex	De Haan

Question 5:

Worksheet Query Builder

```
SELECT department_name
FROM Departments
WHERE department_id = (SELECT department_id FROM Employees WHERE salary = (SELECT MAX(salary) FROM Employees));
```

Script Output x Query Result x

SQL | All Rows Fetched: 1 in 0.007 seconds

DEPARTMENT_NAME
1 Executive

Question 6:

Worksheet

Query Builder

```
SELECT first_name, last_name, salary
FROM Employees e
WHERE salary > (SELECT AVG(salary) FROM Employees WHERE department_id = e.department_id);
```

Script Output x Query Result x

 All Rows Fetched: 28 in 0.01 seconds

	FIRST_NAME	LAST_NAME	SALARY
1	Steven	King	24000
2	Alexander	Hunold	9000
3	Bruce	Ernst	6000
4	Nancy	Greenberg	12008
5	Daniel	Faviet	9000
6	Den	Raphaely	11000
7	Matthew	Weiss	8000
8	Adam	Fripp	8200
9	Payam	Kaufling	7900
10	John	Russell	14000
11	Karen	Partners	13500
12	Alberto	Errazuriz	12000
13	Gerald	Cambrault	11000
14	Eleni	Zlotkey	10500
15	Peter	Tucker	10000
16	David	Bernstein	9500
17	Peter	Hall	9000
18	Janette	King	10000
19	Patrick	Sully	9500
20	Allan	McEwen	9000
21	Clara	Vishney	10500
22	Danielle	Greene	9500
23	Lisa	Ozer	11500
24	Harrison	Bloom	10000
25	Tayler	Fox	9600
26	Ellen	Abel	11000
27	Michael	Hartstein	13000
28	Shelley	Higgins	12008

Question 7:

Worksheet

Query Builder

```
SELECT first_name, last_name, department_id
FROM Employees
WHERE department_id = (SELECT department_id FROM Employees WHERE salary = (SELECT MIN(salary) FROM Employees));
```

Script Output x

Query Result x

 All Rows Fetched: 1 in 0.006 seconds

FIRST_NAME	LAST_NAME	DEPARTMENT_ID
Jennifer	Whalen	10

Question 8:

Worksheet		Query Builder	
		<pre>SELECT first_name, last_name FROM Employees WHERE salary < (SELECT salary FROM Employees WHERE first_name = 'Lex' AND last_name = 'De Haan');</pre>	
Script Output x		Query Result x	
SQL		Fetched 50 rows in 0.005 seconds	
FIRST_NAME	LAST_NAME		
1 Alexander	Hunold		
2 Bruce	Ernst		
3 David	Austin		
4 Valli	Pataballa		
5 Diana	Lorentz		
6 Nancy	Greenberg		
7 Daniel	Faviet		
8 John	Chen		
9 Ismael	Sciarra		
10 Jose Manuel	Urman		
11 Luis	Popp		
12 Den	Raphaely		
13 Alexander	Khoo		
14 Shelli	Baida		
15 Sigal	Tobias		
16 Guy	Himuro		
17 Karen	Colmenares		
18 Matthew	Weiss		
19 Adam	Fripp		
20 Payam	Kaufling		
21 Shanta	Vollman		
22 John	Russell		
23 Karen	Partners		
24 Alberto	Errazuriz		
25 Gerald	Cambrault		
26 Eleni	Zlotkey		
27 Peter	Tucker		
28 David	Bernstein		
29 Peter	Hall		
30 Christopher	Olsen		
31 Nanette	Cambrault		
32 Oliver	Tuvault		
33 Janette	King		
34 Patrick	Sully		
35 Allan	McEwen		

Question 9:

Worksheet		Query Builder	
		<pre>SELECT department_name, manager_id FROM Departments WHERE manager_id = (SELECT manager_id FROM Employees WHERE salary = (SELECT MIN(salary) FROM Employees WHERE job_id LIKE 'MAN%'));</pre>	
Script Output x		Query Result x	
SQL		All Rows Fetched: 0 in 0.006 seconds	
DEPARTM...	MANAGER...		

Question 10:

Worksheet

Query Builder

```
SELECT employee_id, first_name, last_name, salary
FROM Employees
WHERE salary > (SELECT AVG(salary) FROM Employees WHERE hire_date > TO_DATE('01-JAN-2005', 'DD-MON-YYYY'));
```

Script Output x

Query Result x

SQL | All Rows Fetched: 38 in 0.007 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY
1	100	Steven	King	24000
2	101	Neena	Kochhar	17000
3	102	Lex	De Haan	17000
4	103	Alexander	Hunold	9000
5	108	Nancy	Greenberg	12008
6	109	Daniel	Faviet	9000
7	110	John	Chen	8200
8	112	Jose Manuel	Urman	7800
9	114	Den	Raphaely	11000
10	120	Matthew	Weiss	8000
11	121	Adam	Fripp	8200
12	122	Payam	Kaufling	7900
13	145	John	Russell	14000
14	146	Karen	Partners	13500
15	147	Alberto	Errazuriz	12000
16	148	Gerald	Cambrault	11000
17	149	Eleni	Zlotkey	10500
18	150	Peter	Tucker	10000
19	151	David	Bernstein	9500
20	152	Peter	Hall	9000
21	153	Christopher	Olsen	8000
22	156	Janette	King	10000
23	157	Patrick	Sully	9500
24	158	Allan	McEwen	9000
25	159	Lindsey	Smith	8000
26	162	Clara	Vishney	10500
27	163	Danielle	Greene	9500
28	168	Lisa	Ozer	11500
29	169	Harrison	Bloom	10000
30	170	Taylor	Fox	9600
31	174	Ellen	Abel	11000
32	175	Alyssa	Hutton	8800
33	176	Jonathon	Taylor	8600
34	177	Jack	Livingston	8400
35	201	Michael	Hartstein	13000